



Let  $G$  be a graph and  $S \leq \text{Aut}(G)$  be a subgroup of the automorphism group. We define a *graph polyform* to be an induced subgraph of  $G$  where two graph polyforms  $p_1$  and  $p_2$  are equivalent if there exists  $s \in S$  such that the group action sends  $s \cdot p_1 = p_2$ .

If the  $G$  is the infinite grid graph, and the automorphism group consists of all shifts and rotations of  $G$ , then the graph polyforms are the “one-sided” polyominoes.

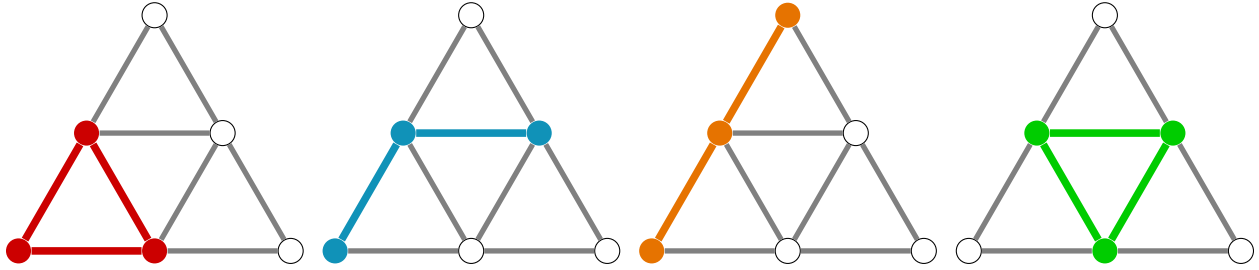


Figure 1: The four 3-vertex graph forms of the triforce graph, up to the six isomorphisms of the graph. There is 1 0-vertex form, 2 1-vertex forms, 2 2-vertex forms, 4 3-vertex forms, 3 4-vertex forms, 2 5-vertex forms, and 1 6-vertex form.

**Question.** Suppose  $G$  is a graph of order  $n$ . When does there exist some  $k$  such that the graph polyforms of size  $k$  partition  $n$ ?

**Related.**

1. Let  $f_G(k)$  be the number of graph polyforms of size  $k$ . When does this sequence fail to be unimodal?
2. Which graphs of order  $n$  admit the most graph polyforms of any size?